

FINAL EXAM

ECO403–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

DECEMBER 1, 2023

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Instructions & Rules: The exam has four problems. Your submission consists of a *single pdf file* that you will submit through Moodle. If you write your solutions in plain paper, scan them with good resolution, and merge them into a single pdf file. Write your solutions neatly and succinctly. Aim for rigorousness as well as efficiency. Any answer that is not supported by an argument will not be valid. Your solutions will be graded based on the personal questions that I will ask about them during the oral exam on December 1. I will evaluate your interview performance based on your accuracy, correctness, coherency, conciseness, as well as exposition.

Please name your submission file according to the following format: **HH.MM_FullName.pdf**, where HH.MM is the time, in 24h format, of your oral interview. Examples:

- If your interview is at 9:30, please name your file 09.30_yourname.pdf.
- If your interview is at 17:30, please name your file 17.30_yourname.pdf.

The rules for the exam are:

1. You can consult your class notes and slides printouts.
2. You can consult any book that I provided as reference.
3. You can consult any other book/reference that you deem suitable, either printed or online.
4. You can perform your *final* calculations with any computer algebra system, but you should be able to explain what you did, i.e., your solutions should include the full development leading to the final step, which then you can perform using the computer.
5. You can consult and work in a group with your ECON403 classmates, but solutions should be written individually. You are not permitted to consult with people outside the class.
6. Any reference that you use, should be acknowledged and cited where applicable. For example, if you consult a webpage or a book, you should give credit and put the proper citation. If you get an idea from a classmate or work in a group, you should acknowledge it.
7. You are not permitted to discuss your interview results with other students.
8. You are not permitted to upload the exam paper to any internet repository or database, except Moodle.

Any deviation from the above set of rules that I detect will be punished with the harshest suitable penalty according to the Student Academic Handbook. Please note that even a category A penalty can involve failing the course at my discretion.

Problem 1: A discrete auction where the seller can restrict the bids.

Consider a seller that can sell an object to one of two bidders. The private values of the two buyers are independent and uniformly distributed over the set $\{0, 1\}$. The seller will organize a first-price auction. The novelty is that now the seller can constrain the bids to be only 0 or x , where $x \in (0, 1]$ is a parameter chosen by the seller. Ties are resolved with a uniform lottery. After the seller publicly announces the value of x , he runs the auction.

1. Represent the first-price auction that the buyers participate in as *Bayesian game*, i.e., represent it as a collection of matrices. Be sure to also include a description of what the players know and their beliefs.
2. What is a pure-strategy in this Bayesian game? What is a mixed-strategy in this game?

For every value of $x \in (0, 1]$, the goal is to find *all* the Bayesian equilibria in the first-price auction above. Note that since the players are symmetric, all equilibrium profiles will be symmetric. Thus, a hint is that anything you do for one player, applies to the other.

You will start with pure-strategy equilibria.

3. For what values of x is the strategy “bid 0 when value 0, bid x when value 1” a strictly dominant strategy?
4. For what values of x is the strategy: “bid 0 always” a strictly dominant strategy?
5. For what values of x are there multiple equilibria and what form they take?

Now you will look for a mixed-strategy equilibrium.

6. For what values of x is there a mixed-strategy equilibrium and what is it?

You will explore now some expected revenue considerations for the seller, who can choose the parameter $x \in (0, 1]$. You will need to look at how expected revenue changes as the value of x moves.

7. Suppose that the seller, when there are multiple equilibria, can lead the players to his *preferred equilibrium*. What is the expected revenue maximizing value of x ?
8. Now suppose that, whenever there are multiple equilibria, the players will choose to play the equilibrium that is the *worst for the seller*. What is now the expected revenue maximizing value of x ?

Problem 2: A Private Values Auction with a seemingly correlated distribution of values.

Consider a private-value auction with two bidders. However, in contrast with the independent distribution of values case from the SIPV model, suppose now that the distribution of values is not independent, but follows the following prior:

$$f(v_1, v_2) = \frac{1}{2} I_{A_1 \cup A_2}(v_1, v_2),$$

where $A_1 = \{(v_1, v_2) \in [0, 1)^2\}$ and $A_2 = \{(v_1, v_2) \in [1, 2]^2\}$, and $I_{A_1 \cup A_2}(v_1, v_2)$ is the indicator function of the set $A_1 \cup A_2$.

1. Give the complete *belief space* description of the situation described above. Be sure to specify what are the signalling functions of the bidders.
2. Compute bidder i 's posterior beliefs after they receive their signal $V_i = v_i$: show that if $v_i \in [0, 1)$ the posterior beliefs are uniformly distributed over $[0, 1]$; and that if $v_i \in [1, 2]$ the posterior beliefs are uniformly distributed over $[1, 2]$.

3. Give a (semi) formal argument that if $(v_1, v_2) \in A_1$, it is common knowledge between the bidders that their values are in A_1 and that their posterior beliefs about the value of their respective opponents are identically distributed as independent uniform random variables over $[0, 1]$. Conclude something similar for bidders whose values are $(v_1, v_2) \in A_2$: it is common knowledge that their values are in A_2 and that their posterior beliefs about the value of their respective opponents are identically distributed as independent uniform random variables over $[1, 2]$. *Hint: you can use the results from items 8 and 9 from the Beliefs Problem List.*
4. Consider a first-price auction. Find the symmetric, monotonically increasing, right-continuous, equilibrium bidding strategy. *Hint: Use 2 and 3 above to simplify the problem to two cases that can be analyzed with the tools from the SIPV model. The equilibrium bidding function will be continuous, except at a point of discontinuity at 1 in which it will jump upwards.*
5. Compute the expected revenue for the first-price auction at the symmetric equilibrium you found in item 4.
6. Consider a second-price auction, and consider its symmetric equilibrium in weakly-dominant strategies. Compute the expected revenue for the second-price auction. Are the expected revenues for the FPA and SPA the same for this exercise?

Problem 3: Interdependent Value Auctions with independent signals

Suppose there are two bidders with signals that are independent and uniformly distributed on the interval $[0, 1]$, and where the valuation functions are given by $v_1(x_1, x_2) = 2x_1 - x_2$, and $v_2(x_1, x_2) = 2x_2 - x_1$.

1. Show that you can define $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ that is symmetric in both arguments and such that $v_1(x_1, x_2) = u(x_1, x_2)$ and $v_2(x_1, x_2) = u(x_2, x_1)$. Use this result to conclude that the auction is one with *common values*.
2. Find the symmetric equilibrium bidding strategy in the first-price sealed bid auction.
3. Find the symmetric equilibrium bidding strategy in the second-price sealed bid auction.
4. Compute the expected revenue associated with each auction format in the equilibria you found above.

Problem 4: Discrete values and Signals

Consider a common-value first-price auction with two bidders. The common value V can take on only two values, 0 or 1. Each bidder i , prior to participating in the auction, receives a signal X_i which also takes only two values, 0 or 1. The joint distribution of V , X_1 and X_2 is

$$\Pr(V = v, X_1 = x_1, X_2 = x_2) = \begin{cases} 2/9 & x_1 = x_2 = v \\ 1/18 & x_1 = x_2 \neq v \\ 1/9 & x_1 \neq x_2 \end{cases}$$

Show that the following strategy constitutes a symmetric equilibrium:

- A bidder with signal $X_i = 0$ bids $E[V|X_1 = 0, X_2 = 0] = \frac{1}{5}$.
- A bidder with signal $X_i = 1$ bids using a mixed strategy $G(b)$ with support over the interval $[\frac{1}{5}, \frac{8}{15}]$ where

$$G(b) = \frac{4}{5} \left(\frac{5b - 1}{4 - 5b} \right)$$